Problem-Solution Fit

OptiCrop — Smart Agricultural Production Optimization Engine

The Problem

Farmers lack a fast, data-driven way to determine which crop is best suited to their soil's actual nutrient levels and local climate, leading to lower yields, wasted resources (fertilizer, water), and avoidable financial loss.

The Solution

OptiCrop trains a classification model on a labelled dataset mapping soil/climate conditions (N, P, K, temperature, humidity, pH, rainfall) to the most suitable crop, and exposes it through a simple web form so any user can get an instant, model-backed recommendation.

Why This Fits

- Directly addresses the root cause (lack of measurable, data-driven guidance) rather than a symptom.
- Requires only the 7 input values a farmer can reasonably obtain or estimate — no specialized hardware needed.
- Achieves high accuracy (99.55% with Random Forest) on a well-established, balanced dataset, making the recommendation trustworthy rather than speculative.
- Scales naturally to research/policy use (Scenario 3) since the same model and dataset can power aggregate insights, not just single predictions.

Validation

The selected approach was validated by comparing four supervised classifiers (Logistic Regression, KNN, Decision Tree, Random Forest) on held-out test data, confirming the model generalizes well rather than overfitting to the training set.